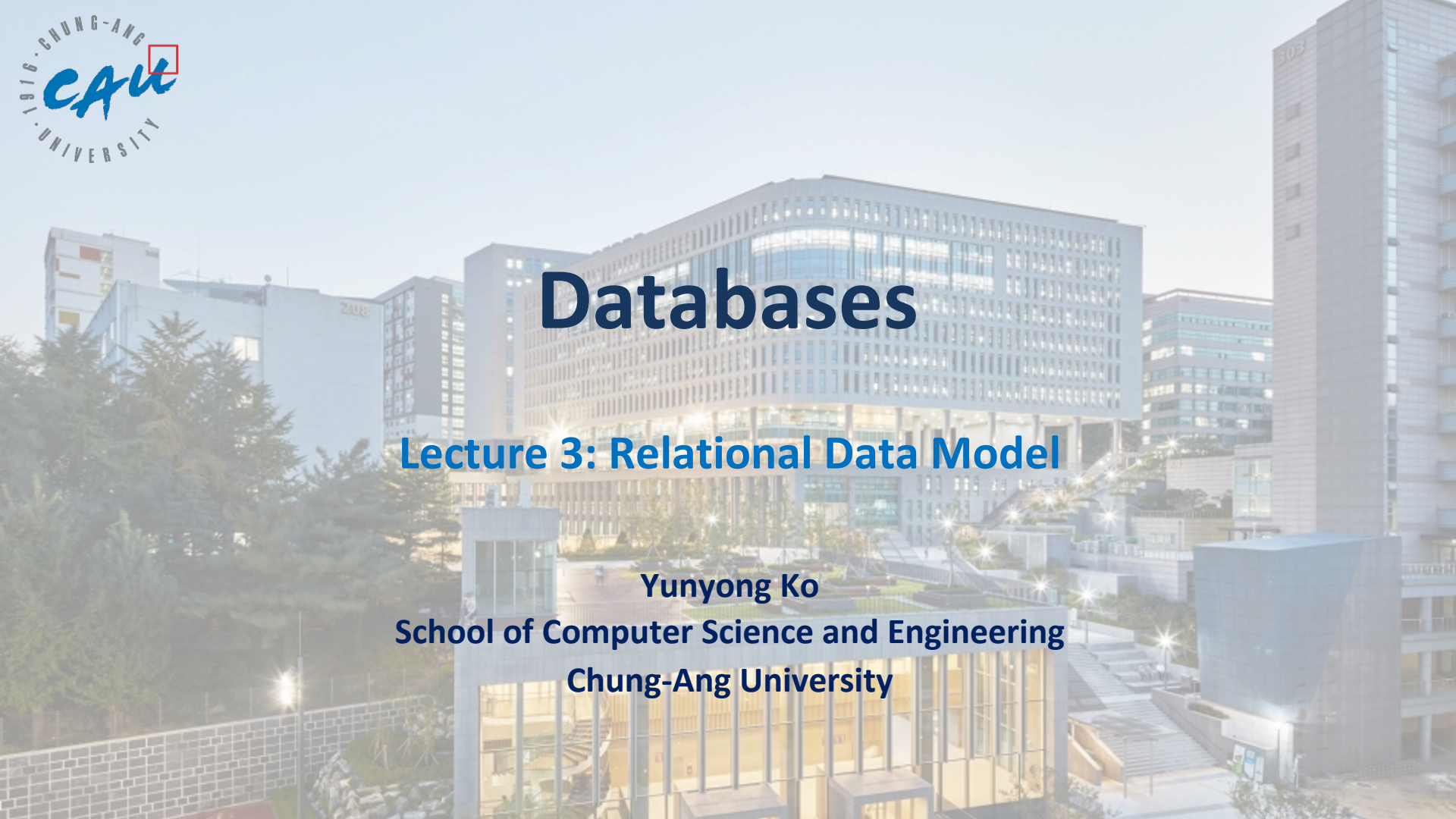


Databases

Lecture 3: Relational Data Model

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Outline

- **Relational Model Concepts**
  - Basic Concepts and Characteristics
  - Relational Model Constraints



- **Operations and Constraint Violations**
  - Basic Operations: Insert, Update, Delete
  - Constraint Violations
- **ER-to-Relational Mapping**
  - Mapping of Entity Types and Multivalued Attributes
  - Mapping of Relationship Types

Recap: Database Design Process

❑ Phase 1: Requirements Collection and Analysis

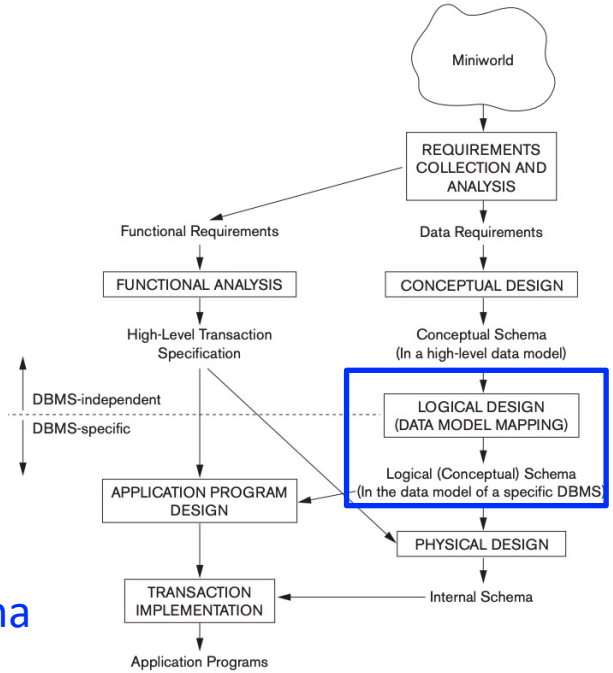
- Characterize the data needs of the prospective DB users
- Collect and analyze data requirements for the database
  - **Output**: a set of users' detailed requirements

❑ Phase 2: Conceptual Design

- Choosing a data model
- Translating the requirements into a conceptual schema

❑ Phase 3: Implementation

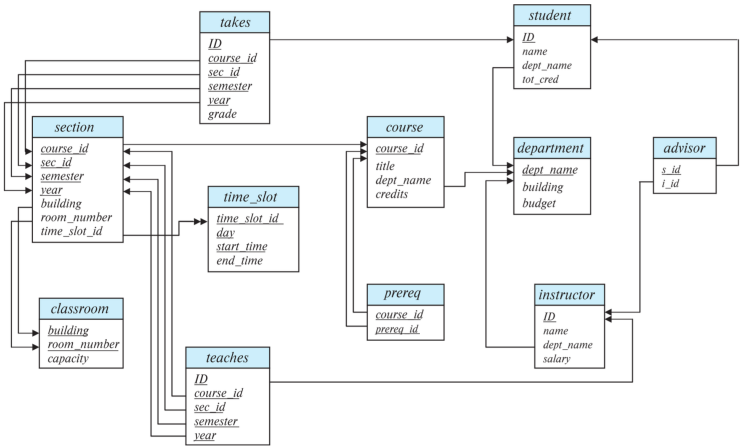
- Implementing the DB based on the conceptual DB schema
  - Logical design: the database schema
  - Physical design: the physical layout of the DB



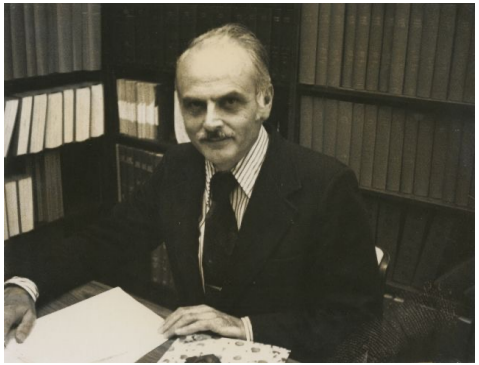
Relational Data Model

What is a Relational Data Model?

- Database design simplifies data representation, making data management efficient
  - Developed by Ted Codd of IBM in 1970
- Using ***relations*** (i.e., *tables*) to represent data, where each relation consists of ***tuples*** (i.e., *rows*) and ***attributes*** (i.e., *column headers*)



Example of Relational Databases



Ted Codd of IBM Research in 1970

Advantages of Relational Model

❑ Simplicity

- Using a flat table structure, making it easy to understand and manage

❑ Data Integrity

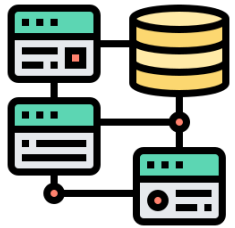
- Ensuring *the data in the database is valid, consistent, and accurate*

❑ Easy Data Manipulation

- Supporting *powerful, high-level data manipulation languages (i.e., SQL)*
  - Users can flexibly manipulate data along with complex conditions

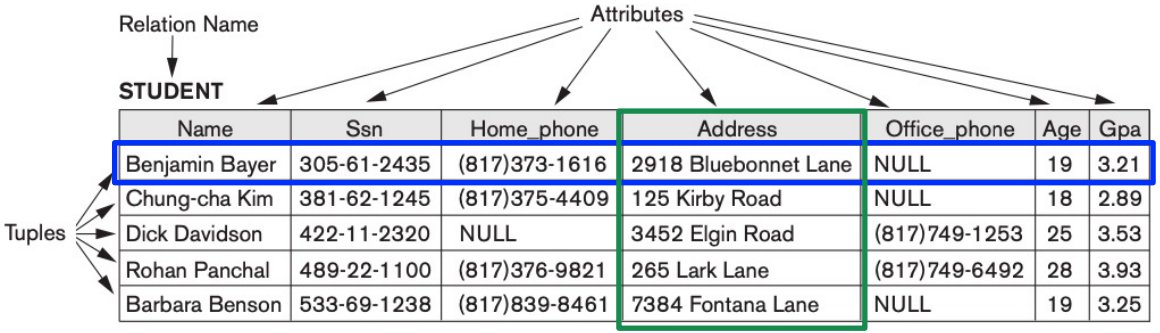
❑ Security

- Offering robust security features,
  - e.g., user authentication, authorization mechanisms, data encryption, and so on



How To Represent a Database?

- **Represent a Database as a Collection of *Relations* (i.e., *Tables*)**
  - A relation consists of a set of ***tuples*** (i.e., *rows*)
    - Each *row* in the table represents a collection of related data values
  - A ***column header*** represents an ***attribute*** defining a property of each object



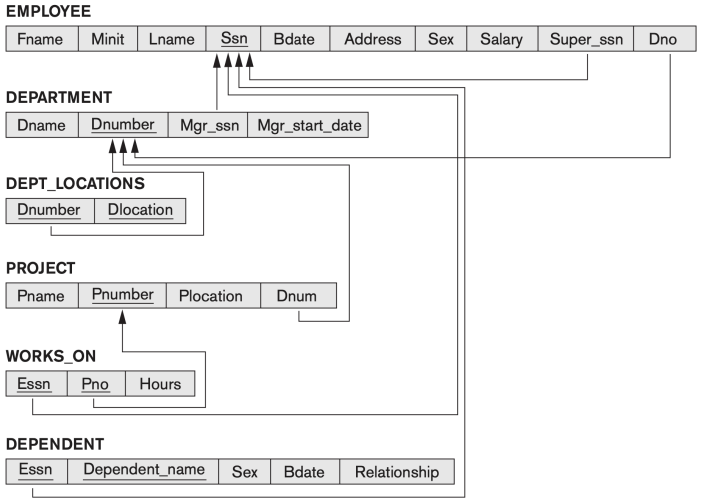
Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

Each row represents facts about a particular entity (student)

Attributes specify how to interpret the data values in each row.

How To Represent a Database? (Cont.)

❑ Example: Company Database



Relational Database **Schema**

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Relational Database **State**

Domain of Attributes

❑ Domain *D*

- A set of ***allowed atomic values*** for each attribute
  - ❑ each value in the domain is indivisible
- Domain = data type + constraints

❑ Domain Examples

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

A **Data type** can be specified for each domain:

The data type for the domain '**Usa_phone_numbers**':
(ddd)ddd-dddd, where d is a numeric digit, and the first three digits form a valid area code

The data type for the domain '**Employee_ages**':
An integer number between 15 and 80

- Usa_phone_numbers. The set of ten-digit phone numbers valid in the United States.
- Local_phone_numbers. The set of seven-digit phone numbers valid within a particular area code in the United States. The use of local phone numbers is quickly becoming obsolete, being replaced by standard ten-digit numbers.
- Social_security_numbers. The set of valid nine-digit Social Security numbers. (This is a unique identifier assigned to each person in the United States for employment, tax, and benefits purposes.)
- Names: The set of character strings that represent names of persons.
- Grade_point_averages. Possible values of computed grade point averages; each must be a real (floating-point) number between 0 and 4.
- Employee_ages. Possible ages of employees in a company; each must be an integer value between 15 and 80.

Relation Schema and Instance (State)

- **Relation Schema:** $R = (A_1, \ldots, A_n)$
  - Made up of a **relation name** and **a list of attributes** $A_1, A_2, \ldots, A_n$
    - e.g., STUDENT $= (\text{Name}, \text{Ssn}, \ldots, \text{Gpa})$
  - The **degree** of a relation: the number of attributes *n* of its relation schema
- **Relation Instance (State):** $r(R)$
  - A set of *n*-tuples (i.e., rows) $r(R) = \{t_1, \ldots, t_m\}$
  - Tuple *t* of Relation *R:* Ordered list of *n* values $t =< v_1, \ldots, v_n >$
    - Each value $v_i (1 \leq i \leq n)$ is an element of dom(A_i) or a special NULL value

Relation Name: STUDENT

Attributes

Tuples

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

Degree of this relation is 7

Characteristics of Relations

Values in Tuples

- Each value in a tuple should be an ***atomic*** value
  - i.e., it is not divisible into components
- *Composite and multivalued attributes are **NOT** allowed*
  - Multivalued attributes must be represented by *separate relations*
  - Composite attributes are represented only by *their simple component attributes*

NULLs in Tuples

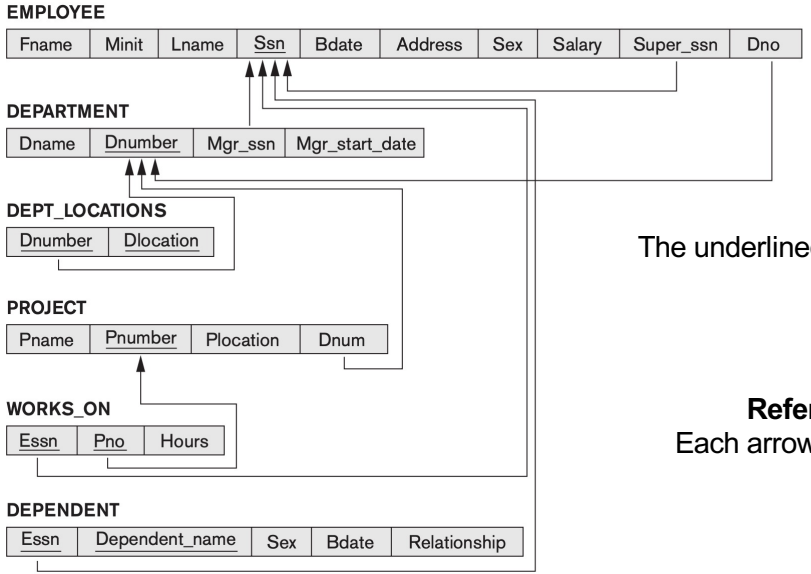
- NULL values are used to represent the values of attributes that may be unknown or may not apply to a tuple
  - Several meanings for NULL values: unknown, unavailable, undefined values, …

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

Relational Database Schema

What is a Relational Database Schema?

- A set of **relation schemas** $S = \{R_1, \ldots, R_m\}$ and a set of ***integrity constraints*** *IC*
  - e.g., COMPANY={EMPLOYEE, DEPARTMENT, DEPT_LOCATIONS, PROJECT, WORKS_ON, DEPEDENT}



Key Constraint:
The underlined attribute represents the primary key

Referential Integrity Constraint:
Each arrow represents the referential relation between two relations

Relational Database Schema for the COMPANY Database

Relational Database State

❑ What is a Relational Database State?

- A set of **relation states** DB $= \{r_1, \ldots, r_m\}$, r_i is a state of R_i
  - ❑ r_i satisfies the integrity constraints specified in IC
  - ❑ A valid DB state should satisfy all the integrity constraints (otherwise, it is invalid)

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

One Possible DB State for the COMPANY Database

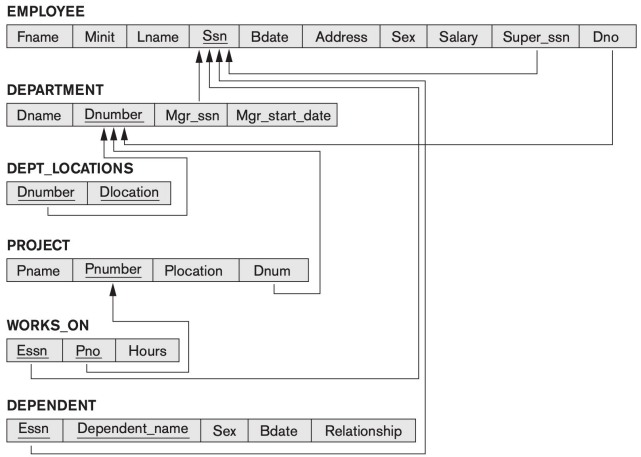
Relational Model Constraints

What are the Relational Model Constraints?

- Specific constraints that must hold for the data
  - There exit many relations, and their tuples are related in various way
  - For data integrity, every relation database state should follow all the constraints (i.e., valid state)
- DBMS should provide **capabilities for defining and enforcing these constraints**

Three Crucial Constraints

- Key Constraints
- Entity integrity Constraints
- Referential integrity Constraints



Key Constraints

What is a *Key Constraint*?

- ***"Two distinct tuples cannot have the identical values for the attributes in SK"***
  - For any two distinct tuples t_1 and t_2 in a relation state r: $t_1[\mathrm{SK}] \neq t_2[\mathrm{SK}]$
- *Superkey (SK)* of Relational Schema *R*
  - A *subset of attributes* in *R* that can uniquely identifies each tuple
  - Every relation has at least one default superkey

Why Key Constraints Required?

- In a relation, tuples must be ***distinct*** by the definition
- How can we identify each tuple uniquely?

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

Key Constraints (Cont.)

❑ *Key* of Relational Schema *R*

- *A superkey* of *R* with the following additional property
  - Removing any attribute ***A*** from ***K*** leaves a set of attributes ***K'*** that is not a superkey of R any more
- It is a ***minimal*** superkey
  - We cannot remove any attributes, the uniqueness constraint holds
  - This *minimality* property is required for a key but optional for a superkey

Superkey: {Ssn, Home_phone}

Removing the attribute "Home_phone"

Key: {Ssn}

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

The value of a key attribute can be used to identify uniquely each tuple

Key Constraints (Cont.)

❑ *Primary and Candidate Keys* of Relational Schema *R*

- A relation schema may have more than one key
  - Each of the keys is called a **candidate key**
- It common to designate one of the candidate keys as the **primary key** of the relation
  - Used to *identify* tuples uniquely in the relation

There are two candidate keys in the CAR relation
License_number is designated as the primary key

Primary key is underlined

CAR

License_number	Engine_serial_number	Make	Model	Year
Texas ABC-739	A69352	Ford	Mustang	02
Florida TVP-347	B43696	Oldsmobile	Cutlass	05
New York MPO-22	X83554	Oldsmobile	Delta	01
California 432-TFY	C43742	Mercedes	190-D	99
California RSK-629	Y82935	Toyota	Camry	04
Texas RSK-629	U028365	Jaguar	XJS	04

Usually, it is better to choose a key with a single attribute or a small number of attributes

Entity Integrity Constraint

❑ What is an *Entity Integrity Constraint*?

- ***"No primary key value can be NULL"***
  - ❑ For every tuple *t* in *r(R)*, $t[PK] \neq NULL$
- The constraint for the primary key (PK) of each relation schema *R*

Office_phone cannot become the key

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

Q. Why entity integrity constraints needed?

If two or more tuples have NULL for their primary keys,
we cannot distinguish them when we try to reference them from other relation

Referential Integrity Constraint

❑ What is a *Referential Integrity Constraint*?

- ***"A value of FK in a tuple t_1 either has a value of PK for some tuple t_2 or is NULL"***
  - *Foreign Key (FK):* a set of attributes in relation schema R_1 **references** R_2
  - i.e., $t_1(FK) = t_2(PK)$ (the tuple t_1 **references** the tuple t_2)
- The FK of R_1 has the same domain(s) as the PK of R_2



EMPLOYEE:
Referencing relation

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Dno: Foreign key (FK)
that references DEPARTMETN



DEPARTMENT:
Referenced relation

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

Dnumber:
PK of DEPARTMENT

Tuple for employee 'John Smith' references the tuple for the 'Research' department,
→ 'John Smith' works for this department

Referential Integrity Constraint (Cont.)

❑ Why is a Referential Integrity Constraint Needed?

- To maintain the consistency among tuples in the two relations
  - ❑ A tuple in one relation referring to another relation ***must refer to an existing tuple*** in that relation



EMPLOYEE:
Referencing relation



DEPARTMENT:
Referenced relation

The **Dno of EMPLOYEE** gives the department number for which each employee works

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	3
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

Q. What if the value of the Dno is 3?

?

We cannot interpret the meaning of this attribute value

Exercises: Referential Integrity Constraints

❑ Exercise 1

- Consider the following relations for a database that keeps track of business trips of salespersons in a sales office:

SALESPERSON(Ssn, Name, Start_year, Dept_no)

TRIP(Ssn, From_city, To_city, Departure_date, Return_date, Trip_id)

EXPENSE(Trip_id, Account#, Amount)



- A trip can be charged to one or more accounts
- Specify all foreign keys for this schema

Exercises: Referential Integrity Constraints

❑ Exercise 2

- Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:

STUDENT(Ssn, Name, Major, Bdate)
COURSE(Course#, Cname, Dept)
ENROLL(Ssn, Course#, Quarter, Grade)
BOOK_ADOPTION(Course#, Quarter, Book_isbn)
TEXT(Book_isbn, Book_title, Publisher, Author)



- Specify all foreign keys for this schema

Outline

- Relational Model Concepts
  - Basic Concepts and Characteristics
  - Relational Model Constraints
- **Operations and Constraint Violations**
  - Basic Operations: Insert, Update, Delete
  - Constraint Violations
- **ER-to-Relational Mapping**
  - Mapping of Entity Types and Multivalued Attributes
  - Mapping of Relationship Types



Operations of Relational Model

❑ Two Types of Operations

- **Retrieval**: to retrieve information from the database
- **Update**: to change the states of relations in the database

❑ Basic Update Operations

- **Insert**: insert one or more new tuples into a relation
- **Delete**: remove tuples from a relation
- **Update** (or **Modify**): change attribute values in existing tuples





Whenever these operations are applied, some constraints may be violated

Basic Update Operation: Insert

❑ Insert Operation

- Add a single tuple (row) to a relation (table)
  - ❑ Need to specify the relation name and a list of values for the tuple

Operation:
Insert <'Cecilia', 'F', 'Kolonsky', '677678989', '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, NULL, 4> into EMPLOYEE.

Insert a tuple with all attribute values
The values should be listed in the same order

Q. What kinds of constraints can be violated by Insert operations?

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

Basic Update Operation: Insert (Cont.)

❑ Constraint Violations by the Insert Operation

- **Key constraint**: a key value in the new tuple already exist in another tuple
- **Entity constraint**: any part of the PK of the new tuple is NULL

Operation:
Insert <'Alicia', 'J', 'Zelaya', '999887777', '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, '987654321', 4> into EMPLOYEE.

→ This operation violates **the key constraint**

Operation:
Insert <'Cecilia', 'F', 'Kolonsky', NULL, '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, NULL, 4> into EMPLOYEE.

→ This operation violates **the entity integrity constraint**

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

Basic Update Operation: Insert (Cont.)

❑ Constraint Violations by the Insert Operation

- **Referential integrity constraint**: referenced tuples do not exist

Operation:
Insert <'Cecilia', 'F', 'Kolonsky', '677678989', '1960-04-05', '6357 Windswept, Katy, TX', F, 28000, '987654321', 7> into EMPLOYEE.

→ This operation violates **the referential integrity constraint**

Operation:
Insert <'Cecilia', 'F', 'Kolonsky', '677678989', '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, NULL, 4> into EMPLOYEE.

→ This operation satisfies all constraints

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

There is no tuple with Dnumber=7

Basic Update Operation: Delete

❑ Delete Operation

- Remove a tuple from a relation
  - ❑ Need to specify the relation name and the values of PK attributes

Operation:
Delete the WORKS_ON tuple with Essn = '999887777' and Pno = 10.

Q. What kinds of constraints can be violated by Delete operations?

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

Basic Update Operation: Delete (Cont.)

❑ Constraint Violations by the Delete Operation

- **Referential integrity constraint**: the deleted tuple is referenced from other tuples

Operation:
Delete the EMPLOYEE tuple with Ssn = '999887777'

Two tuples in WORKS_ON refers to this tuple
→ Violate **the referential integrity constraint**

Operation:
Delete the EMPLOYEE tuple with Ssn = '333445555'

Four tuples in WORKS_ON refers to this tuple
Two tuples in EMPLOYEE refers to this tuple
→ Violate **the referential integrity constraint**

Operation:
Delete the WORKS_ON tuple with Essn = '999887777' and Pno = 10.

No tuples refers to this tuple
→ This operation satisfies the referential integrity constraint

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

Basic Update Operation: Update

❑ Update (Modify) Operation

- Modify the values of one or more attributes
  - ❑ Need to specify the attributes being updated and their values

Operation:
Update the salary of the EMPLOYEE tuple with Ssn = '999887777' to 28000.

Operation:
Update the Dno of the EMPLOYEE tuple with Ssn = '999887777' to 1.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Q. What kinds of constraints can be violated by Update operations?

Basic Update Operation: Update (Cont.)

❑ UPDATE Examples Violating Constraints

Operation:
Update the Dno of the EMPLOYEE tuple with Ssn = '999887777' to 7.

→ This operation violates **the referential integrity constraint**

Operation:
Update the Ssn of the EMPLOYEE tuple with Ssn = '999887777' to '987654321'

→ This operation violates
both **the key and the referential integrity constraints**

Only updating the attributes of PK or FK can cause the problems

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

Outline

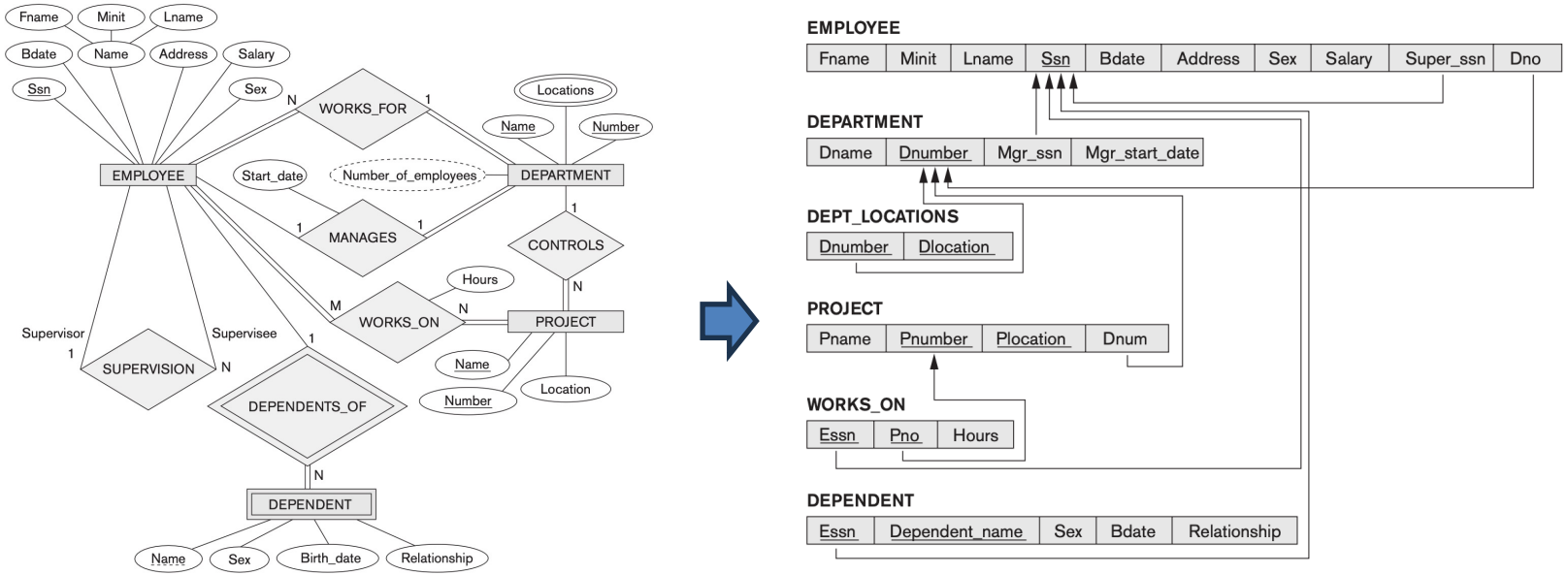
- Relational Model Concepts
  - Basic Concepts and Characteristics
  - Relational Model Constraints
- Operations and Constraint Violations
  - Basic Operations: Insert, Update, Delete
  - Constraint Violations
- **ER-to-Relational Mapping**
  - Mapping of Entity Types and Multivalued Attributes
  - Mapping of Relationship Types



Relational Database Design

❑ ER-to-Relational Mapping

- Let's design a relational database schema based on a conceptual schema design



ER-to-Relational Mapping

Process of ER-to-Relational Mapping

❑ Mapping of Entity Types

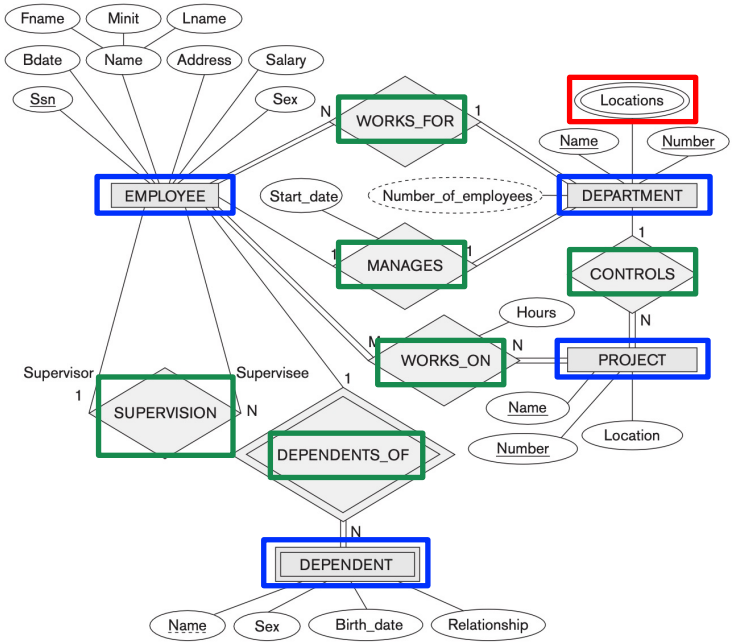
- Regular and weak entity types

❑ Mapping of Multivalued Attributes

- Multivalued attributes are not allowed

❑ Mapping of Relationship Types

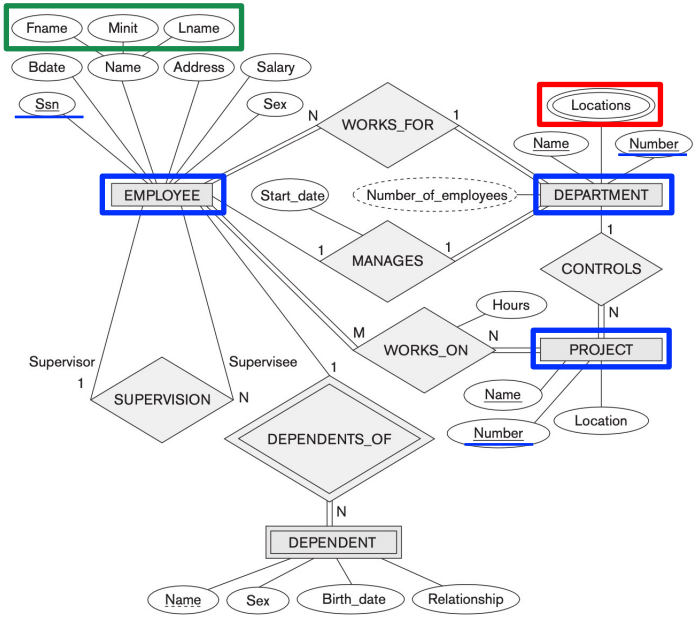
- Binary relationship types: 1:1, 1:N, M:N
- N-ary relationship types (e.g., ternary)



Process 1: Regular Entity Types

❑ Mapping a Regular Entity Type

- (1) Create a relation *R* that includes all the simple attributes of the entity type *E*
  - ❑ For a composite attribute, include only the simple component attributes of the composite attribute
- (2) Designate one of the key attributes of *E* as the primary key for *R*



EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary

DEPARTMENT

?

Dname	Dnumber	Locations

How do we map a multivalued attribute?

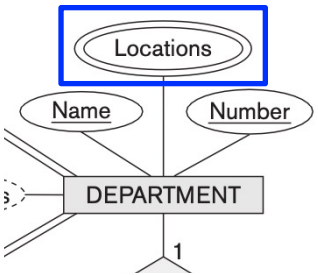
PROJECT

Pname	Pnumber	Plocation

Process 1-2: Multivalued Attributes

❑ Mapping a Multivalued Attribute

- (1) Create a *separate* relation *R* for each multivalued attribute A
- (2) Includes the attribute *A* and the primary key of the original relation
  - ❑ Designate the original relation's primary key as a **foreign key** in R
  - ❑ The key attribute value of the entity (FK) is repeated
- (3) Set the combination of *A* and *K* as the primary key



DEPARTMENT

Dname	Dnumber

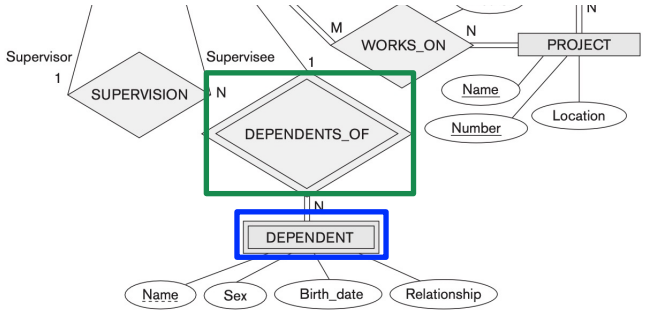
DEPT_LOCATIONS

Dnumber	Dlocation
5	Seoul
5	Incheon
5	Suwon
3	Seoul
3	Daejeon

Process 2: Weak Entity Types

❑ Mapping a Weak Entity Type

- (1) Create a relation *R* including all the simple attributes of the weak entity type *W*
  - ❑ For a composite attribute, include only the simple component attributes of the composite attribute
- (2) Include the primary key of the owner entity type as a ***foreign key***
  - ❑ This takes care of mapping the identifying relationship type of *W*
- (3) Set the combination of the owner's PK and the partial key as the primary key



DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship

Mapping the identifying relationship type

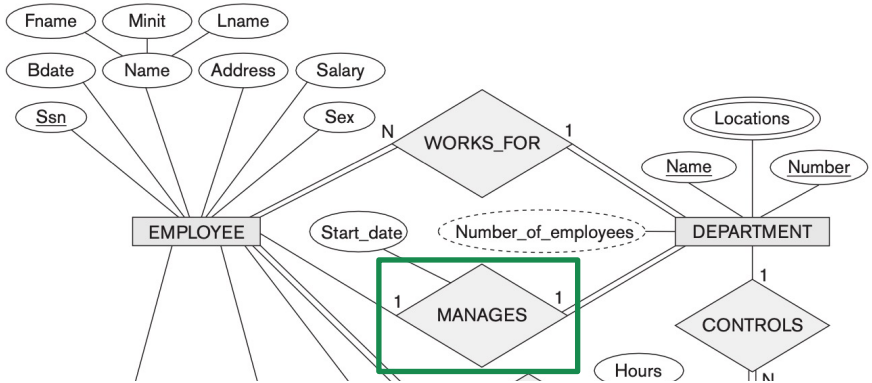
EMPLOYEE

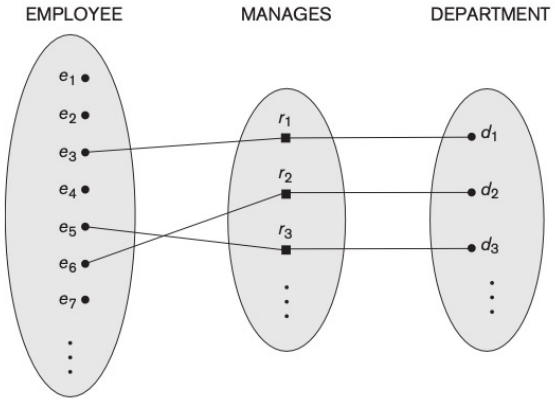
Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary

Process 3-1: Binary Relationship Types (1:1)

❑ Three Possible Approaches

- ***Foreign key approach*** (*most frequently used and useful*)
- Merged relation approach
- Cross-reference (relationship relation) approach

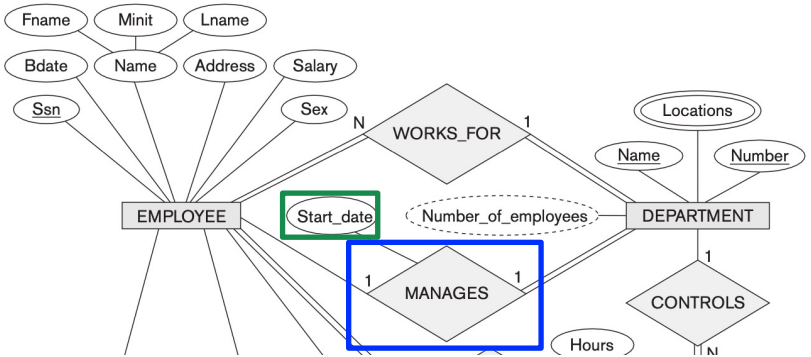




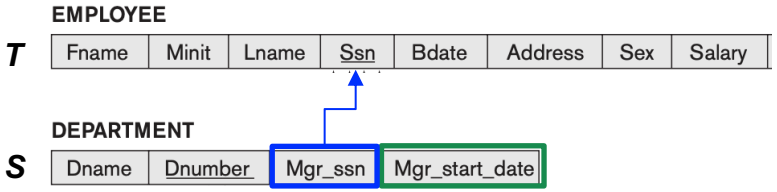
Process 3-1: Binary Relationship Types (1:1)

❑ Foreign Key Approach

- (1) Identify the participating entity types *S* and *T* of the relationship type R
- (2) Choose one of the relations and include the PK of the other one as an FK
  - ❑ It is better to choose an entity type with *total participation in* the role of *S* (Why?)
- (3) Include simple attributes of the relationship type *R* as attributes of *S*



Relationship type attributes can be migrated into one of any participating entities

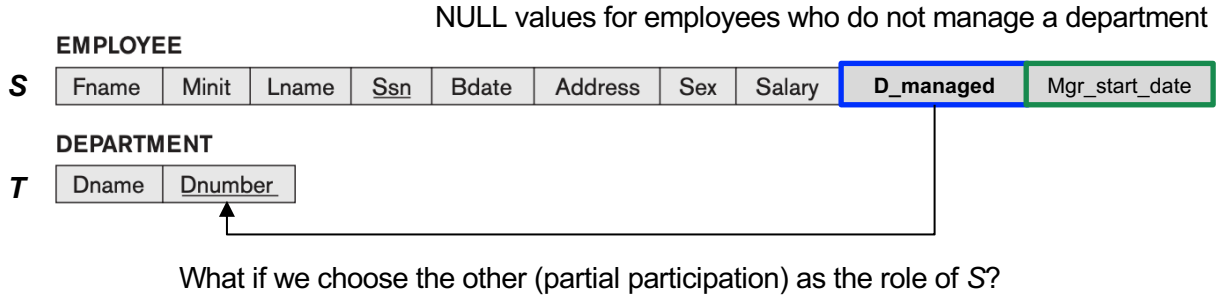


The entity type DEPARTMENT is chosen as the role of *S*

Process 3-1: Binary Relationship Types (1:1)

❑ Foreign Key Approach

- (1) Identify the participating entity types *S* and *T* of the relationship type R
- (2) Choose one of the relations and include the PK of the other one as an FK
  - ❑ It is better to choose an entity type with *total participation* in the role of *S* (Why?)
- (3) Include simple attributes of the relationship type *R* as attributes of *S*

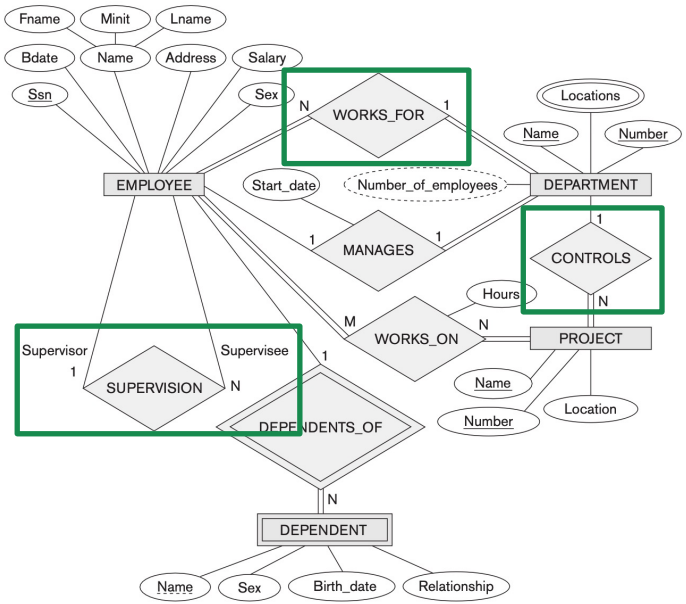


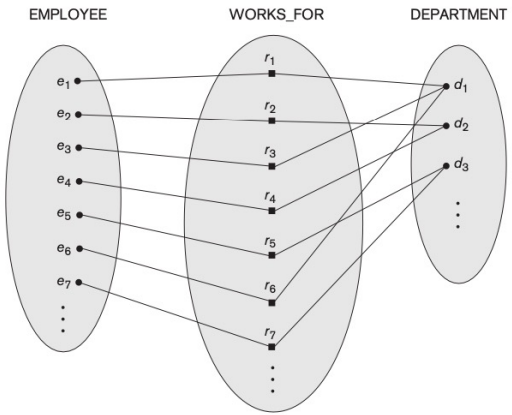
What if we choose the other (partial participation) as the role of *S*?

Process 3-2: Binary Relationship Types (1:N)

❑ Two Possible Approaches

- ***Foreign key approach*** (preferred as it reduces the number of relations)
- Cross-reference (relationship relation) approach

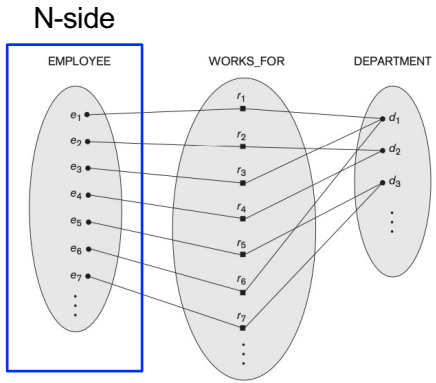




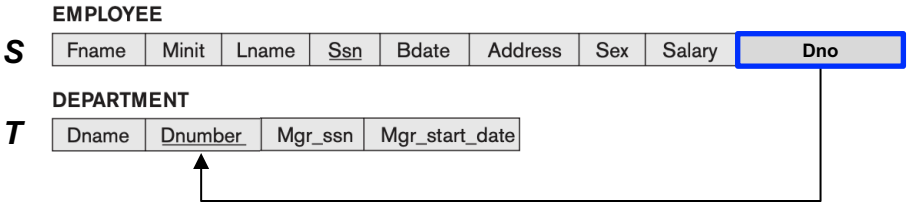
Process 3-2: Binary Relationship Types (1:N)

❑ Foreign Key Approach

- (1) Identify the relation *S* representing the ***N-side*** entity type of the relationship type
- (2) Include the primary key of the other relation *T* as a foreign key in *S*
- (3) Include simple attributes of the relationship type as attributes of *S*



Each entity instance on the N-side is related to at most one entity instance on the 1-side



Process 3-2: Binary Relationship Types (1:N)

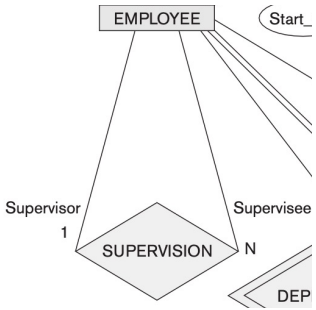
❑ Foreign Key Approach

- (1) Identify the relation *S* representing the ***N-side*** entity type of the relationship type
- (2) Include the primary key of the other relation *T* as a foreign key in *S*
- (3) Include simple attributes of the relationship type as attributes of *S*

SUPERVISION relationship type

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Dno	**Super_ssn**



CONTROLS relationship type

DEPARTMENT

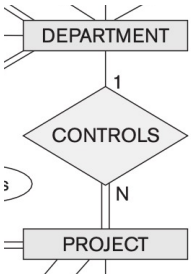
T

Dname	Dnumber	Mgr_ssn	Mgr_start_date

PROJECT

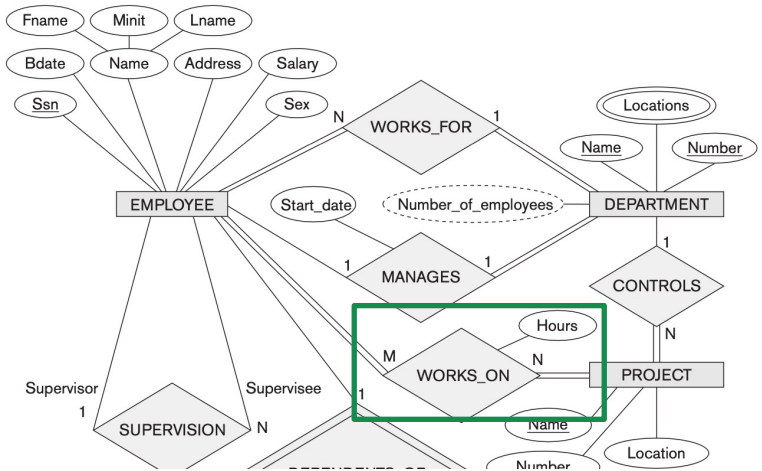
S

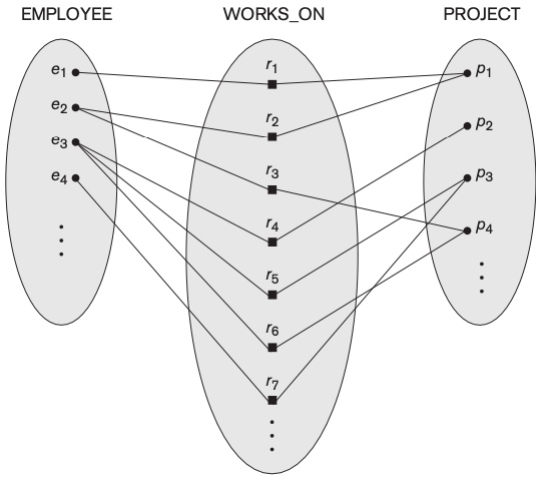
Pname	Pnumber	Plocation	**Dnum**



Process 3-3: Binary Relationship Types (M:N)

- **Only One Possible Approach**
  - **Cross-reference (relationship relation) approach**

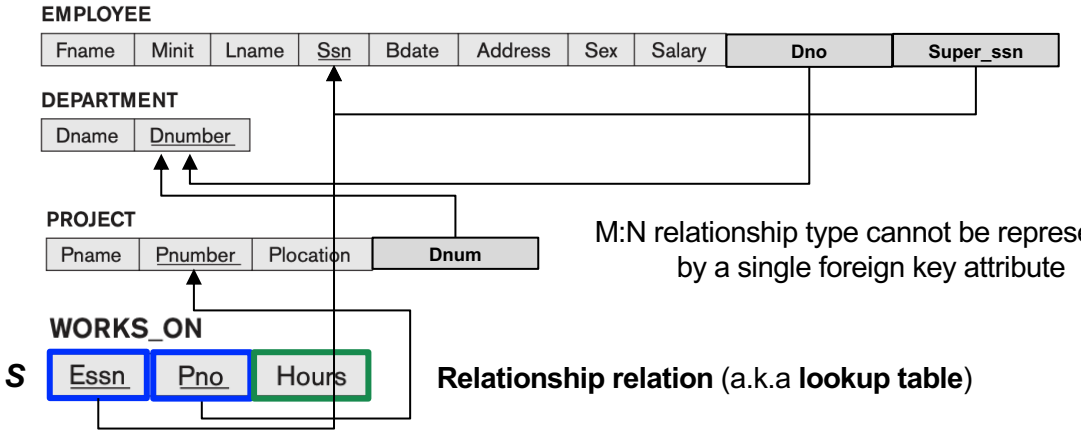




Process 3-3: Binary Relationship Types (M:N)

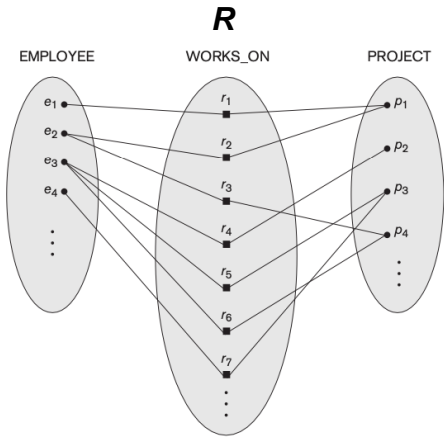
❑ Cross-Reference Approach

- (1) Create a separate relation *S* to represent the relationship type *R*
- (2) Include *the primary keys of the relations representing the participating entity types as foreign keys* in *S*
- (3) Set the combination of all the foreign keys as the primary key
- (4) Include simple attributes of M:N relationship type as attributes of *S*



M:N relationship type cannot be represented by a single foreign key attribute

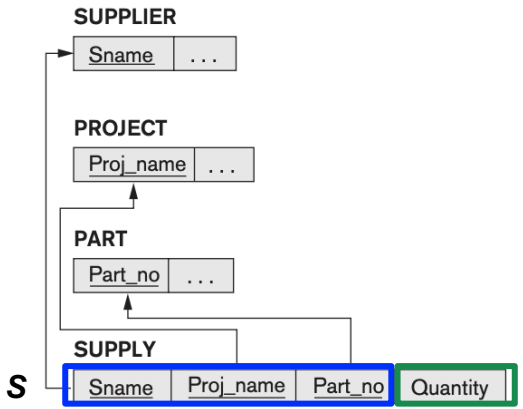
Relationship relation (a.k.a **lookup table**)

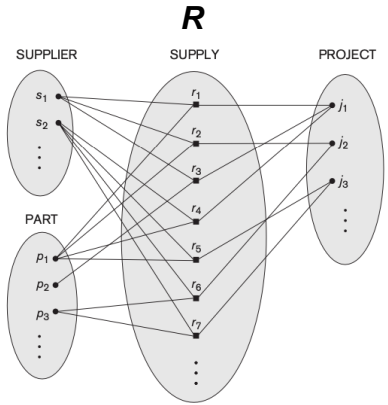


Process 4: N-ary Relationship Types

❑ Cross-Reference Approach

- (1) Create a separate relation *S* to represent the relationship type *R*
- (2) Include *the primary keys of the relations representing the participating entity types as foreign keys* in *S*
- (3) Set the combination of all the foreign keys as the primary key
- (4) Include simple attributes of n-ary relationship type as attributes of *S*

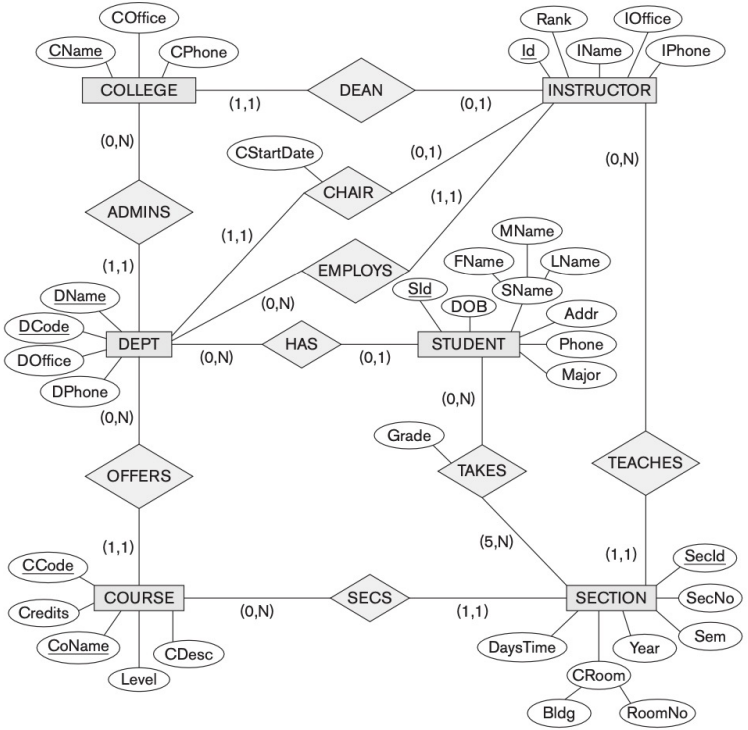




Exercises: ER-to-Relational Mapping

❑ Exercise 3:

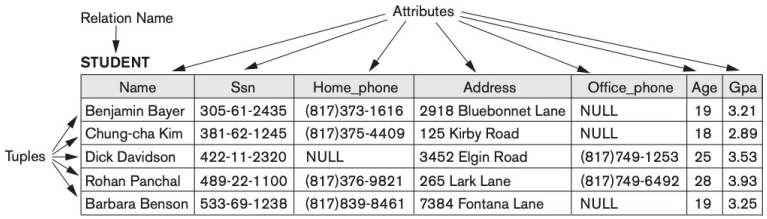
- Map the ER schema for UNIVERSITY database into a relational database schema



Takeaways: Relational Data Model

❑ Relational Model Concepts

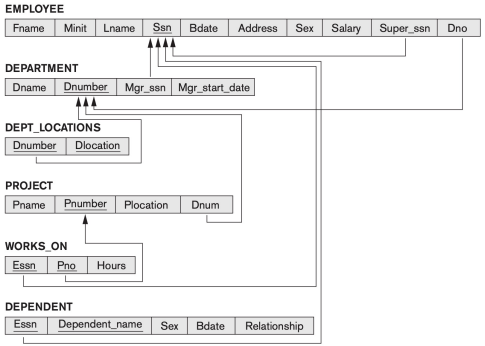
- Relations, tuples, attributes, domains
- Characteristics of Relations
- Relational database schema and states



Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

❑ Relational Model Constraints

- Key Constraints
  - *Two distinct tuples cannot have the identical values for the key attributes*
- Entity Integrity Constraints
  - *No primary key value can be NULL*
- Referential Integrity Constraints
  - *A value of FK in a tuple either has a value of PK for some tuple or is NULL*



Takeaways: Relational Data Model (Cont.)

❑ Basic Update Operations

- **Insert**: insert one or more new tuples into a relation
- **Delete**: remove tuples from a relation
- **Update** (or **Modify**): change attribute values in existing tuples

❑ Constraint Violations

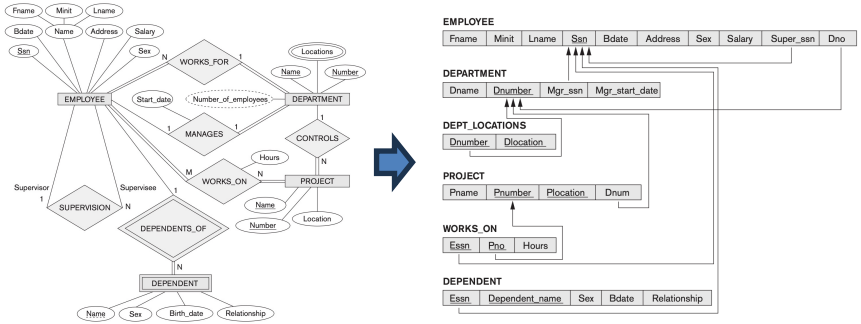
- **Insert**: may violate any of constraints (*key, entity integrity, and referential integrity*)
- **Delete**: may violate the *referential integrity constraint*
- **Update**: may violate the *key and referential integrity constraint*
  - Only updating the attributes of PK or FK can cause the problems





Takeaways: ER-to-Relational Mapping

❑ Correspondence between ER and Relational Models



ER MODEL	RELATIONAL MODEL
Entity type	*Entity* relation
1:1 or 1:N relationship type	Foreign key (or *relationship* relation)
M:N relationship type	*Relationship* relation and *two* foreign keys
n-ary relationship type	*Relationship* relation and *n* foreign keys
Simple attribute	Attribute
Composite attribute	Set of simple component attributes
Multivalued attribute	Relation and foreign key
Value set	Domain
Key attribute	Primary (or secondary) key

Q&A:
Relational Data Model